King Fahad Central Hospital

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Hep

VTE Prophylaxis & Selected Therapeutic Anticoagulation — Aligned with AHA/ASA Guidelines for Early Management of Acute Ischemic Stroke

⚠HIGH-ALERT MEDICATION — This is a generic template for institutional customization only. Heparin (particularly IV unfractionated heparin) is a high-alert medication with narrow therapeutic margin. Dose calculation and pump programming MUST undergo independent double-check by two clinicians per institutional high-alert medication policy. This document must be reviewed and approved by Pharmacy & Therapeutics and the Stroke Committee, and checked against the current institutional heparin nomogram and AHA/ASA guideline version, before clinical use. It does not replace individualized physician judgment.

1. Guiding Principle

Routine urgent full-dose (therapeutic) anticoagulation is NOT recommended for the general treatment of acute ischemic stroke, including for atrial fibrillation, and has not been shown to improve outcomes while increasing hemorrhage risk. Heparin's standardized, protocol-driven role in acute stroke care is venous thromboembolism (VTE) prophylaxis at prophylactic dosing. Full-dose (therapeutic) heparin is reserved for selected, physician-specified indications and is never a default or protocol-driven order.

2. Indications

2.1 VTE Prophylaxis (standardized, protocol-driven)

☐ **Immobile hospitalized stroke patients, once intracranial hemorrhage has been excluded on imaging and any thrombolysis-related hold period has passed**

☐ Used in conjunction with mechanical prophylaxis (sequential compression devices) per institutional Acute Ischemic Stroke admission order set

2.2 Therapeutic (Full-Dose) Anticoagulation — Selected Indications Only, Physician-Specified

These indications require an individualized physician order — they are not part of a standardized, automatic stroke order set.

☐ Cerebral venous sinus thrombosis (CVST) — anticoagulation is generally indicated even in the presence of associated hemorrhagic venous infarction, per physician/neurology judgment

☐ High-risk cardioembolic source requiring urgent anticoagulation per physician judgment (e.g., mechanical heart valve, selected left ventricular thrombus)

☐ Selected symptomatic extracranial cervical artery dissection, per physician judgment (evidence comparing anticoagulation vs. antiplatelet therapy is mixed; individualize)

☐ Other physician-specified indication with documented rationale: ___________

☐ **Atrial fibrillation alone, in the absence of one of the above specific indications, is NOT a standard indication for urgent IV heparin in acute ischemic stroke — routine oral anticoagulation timing is addressed separately per institutional antithrombotic guidance**

3. Contraindications / Cautions

☐ Intracranial hemorrhage on imaging or high suspicion of hemorrhagic transformation

☐ **Within 24 hours of IV thrombolysis, until repeat imaging excludes hemorrhage**

☐ Recent mechanical thrombectomy — hold until access-site hemostasis confirmed and post-procedure imaging excludes hemorrhage, per institutional Post-Thrombectomy Care protocol

☐ Active serious bleeding or known bleeding diathesis

☐ Severe uncontrolled hypertension

☐ Severe thrombocytopenia or history of heparin-induced thrombocytopenia (HIT)

☐ Recent major surgery, trauma, or planned invasive procedure — coordinate timing with proceduralist

☐ Severe renal impairment — dose adjustment/agent selection per pharmacy input

4. Pre-Administration Safety Checks (High-Alert Medication)

☐ **Non-contrast head CT reviewed — no hemorrhage present**

☐ Actual measured body weight used for any weight-based dosing (never estimated)

☐ Baseline aPTT, platelet count, and renal function obtained before starting therapeutic-dose heparin

☐ Indication and physician order clearly documented, specifying prophylactic vs. therapeutic intent

☐ **For therapeutic dosing: independent double-check of dose calculation and pump programming by two clinicians before administration**

☐ Review of home/recent anticoagulant or antiplatelet use, and timing relative to any thrombolysis given

5. VTE Prophylaxis Dosing

Agent	Typical Dosing (confirm against institutional formulary)
Unfractionated heparin (subcutaneous)	5,000 units subcutaneously every 8–12 hours
Low molecular weight heparin (e.g., enoxaparin)	Per institutional formulary and renal-function–adjusted dosing

☐ Initiate once hemorrhage excluded and any post-thrombolysis hold period has elapsed, generally by hospital day 2 per core measure timing

☐ Continue mechanical prophylaxis (SCDs) concurrently, especially before pharmacologic prophylaxis is started or if pharmacologic prophylaxis is held

6. Therapeutic (Full-Dose) IV Heparin Dosing — When Specifically Ordered

Use the current institution-specific weight-based heparin nomogram — the figures below illustrate a common structure only and must be replaced with the approved local nomogram before use.

Component	Illustrative Structure (replace with institutional nomogram)
Loading bolus	Per physician order and indication — omit or reduce bolus in patients with recent stroke given bleeding risk; individualize
Initial infusion rate	Weight-based (units/kg/hr) per institutional nomogram
Target aPTT / anti-Xa	Per institutional nomogram (e.g., aPTT 1.5–2.5× control, or institution-specific anti-Xa target)
Rate titration	Per institutional nomogram based on aPTT/anti-Xa results

☐ **Two-clinician independent double-check of weight-based bolus/infusion calculation and pump programming before starting or adjusting the infusion**

☐ Use a dedicated IV line per institutional high-alert medication infusion policy

7. Monitoring

☐ For therapeutic IV heparin: aPTT (or anti-Xa per institutional protocol) drawn per nomogram — commonly every 6 hours until therapeutic and stable, then per institutional interval

☐ Platelet count obtained at baseline and monitored per institutional HIT-surveillance protocol during therapeutic anticoagulation

☐ Neuro checks and vital signs per institutional stroke monitoring schedule; monitor closely for signs of new or worsening bleeding, including neurologic decline suggestive of hemorrhagic transformation

☐ Monitor for signs of systemic bleeding: hematuria, melena, unexplained hypotension, access-site or injection-site bleeding

☐ Renal function monitored periodically, particularly with LMWH or prolonged therapy

8. Heparin-Induced Thrombocytopenia (HIT)

☐ Maintain clinical suspicion for HIT with unexplained platelet count fall (commonly defined as >50% drop from baseline), typically occurring 5–10 days after heparin initiation, or more rapidly with recent prior heparin exposure

☐ If HIT suspected: stop all heparin products (including line flushes), notify physician immediately, and evaluate per institutional HIT diagnostic/management pathway (e.g., 4Ts score, laboratory HIT testing)

☐ Alternative non-heparin anticoagulant considered per physician/hematology guidance if anticoagulation still required and HIT confirmed or highly suspected

9. Reversal — Major Bleeding on Heparin

☐ Stop heparin infusion immediately

☐ Notify physician/stroke team immediately; obtain emergent imaging if intracranial bleeding suspected

☐ Protamine sulfate reversal considered for significant heparin-associated bleeding, dosed per current institutional protocol/pharmacy guidance based on estimated circulating heparin dose and time since last dose — verify exact dosing with pharmacy at the time of the event given the narrow safety margin of protamine itself

☐ Obtain STAT labs: CBC, aPTT, platelet count, type and crossmatch as indicated

☐ Manage per institutional major hemorrhage/reversal protocol; involve hematology and/or neurosurgery as indicated

10. Timing Relative to Reperfusion Therapy

☐ **Do not initiate therapeutic heparin within 24 hours of IV thrombolysis, and only after repeat imaging excludes hemorrhage**

☐ Following mechanical thrombectomy, hold anticoagulation until access-site hemostasis is confirmed and post-procedure imaging excludes hemorrhage, per institutional Post-Thrombectomy Care protocol

☐ Coordinate timing with any planned procedure (e.g., PEG placement, tracheostomy) — hold per proceduralist guidance

11. Documentation

Patient Name: ______________________	**Actual Body Weight:** ______ kg
MRN: ____________________	**Indication:** VTE prophylaxis / Therapeutic (specify): ______

☐ Indication for heparin therapy documented (prophylactic vs. therapeutic, and specific rationale if therapeutic)

☐ Independent double-check documented for any weight-based/therapeutic dosing

☐ Baseline and follow-up labs (aPTT, platelet count) documented

☐ Any bleeding events, HIT concerns, or dose adjustments documented with time and action taken

Physician Order & Verification

Ordering Physician Signature: ________________________ Date/Time: __________________

Preparer/Administering RN: _______________________ Independent Verifier: _______________

References: 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates (recommendations against routine urgent anticoagulation in acute ischemic stroke; VTE prophylaxis recommendations); AHA/ASA Get With The Guidelines–Stroke VTE prophylaxis core measure; institutional Pharmacy & Therapeutics heparin nomogram and high-alert medication policy. This template must be re-verified against the current institutional nomogram, current manufacturer prescribing information, and the current guideline version at the time of institutional adoption.